

Data Processing

Learning Objectives

- Understand the input-process-output cycle
- Identify different types of data processing
- Explain how computers manipulate data automatically

Engage (5-7 minutes)

Think of a flour mill: farmer brings wheat (input), mill grinds it (process), and flour comes out (output). Computers work the same way with data. Today we explore how computers process information step by step.

Explore (10-12 minutes)

List five daily activities that follow input-process-output: cooking a meal, doing homework, washing clothes. For each, identify the input, the process, and the output. Share your list with the class.

Explain (10-12 minutes)

Data processing follows the IPO cycle: Input collects raw data, Process transforms it using instructions, Output delivers results. Computers perform arithmetic, comparison, sorting, and searching operations on data automatically and at incredible speed.

Elaborate (8-10 minutes)

A shopkeeper enters item prices (input), computer calculates total with tax (process), and prints a bill (output). Design a flowchart for a school marks system: input marks, calculate grades, output report cards. Draw it step by step.

Evaluate (5-8 minutes)

Exit ticket: Define the IPO cycle with one real-world example. Name two types of processing operations a computer can perform.